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Poster Project Outline: Gene Expression in Short Term Spheroid Culture of High-Grade Serious Ovarian Cancer (HGSOC) Patients

Introduction

Of all ovarian cancers, around 85-90% of them are epithelial ovarian cancer. (Kim et al. 2018) High-Grade Serous Ovarian Cancer (HGSOC) is the most common subtype of the more aggressive type II epithelial ovarian cancer, accounting for 70% of all epithelial ovarian cancer. (Kim et al. 2018) Genetically, HGSOC can be characterized by its high genomic instability, especially in tumour suppressor TP53 as more than 96% of HGSOC samples collected contained some form of somatic TP53 mutations. (Lisio et al. 2019) Unfortunately, the disease is often diagnosed at advanced stages where cancer cells have been metastasized since in early stages symptoms can be vague and not easily detectable. (Kim et al. 2018, Kordowitzki et al. 2025) Due to its later diagnosis, HGSOC is responsible for 73.9% of epithelial ovarian cancer deaths. (Kim et al. 2018) Effective treatment is necessary to reduce the mortality of HGSOC.

To increase treatment efficiency, reduce side effects, and improve patient survival, personalized medicine is often used for cancer patients. Patient-derived organoids (PDOs) can be used to develop genomic models to better understand the patient's disease, or to test drug responses ex vivo to identify the most effective treatment for the patient. (Wensink et al. 2021, Vias et al. 2025)

Knowledge Gaps

The formation of multicellular spheroids is a massive complicating factor in treating complex cancers, conferring resistance to conventional chemotherapies and promoting recurrence. Current understanding of the development and behaviour of spheroids remains understudied. Spheroids also have the potential to more accurately model *in vivo* tumours to identify more effective, individualized treatments for aggressive and late-stage cancers. Our goal is to better understand the expression patterns of these short-term spheroid cultures to address some of these shortcomings.

Description of Data

The dataset consists of eight individual files of culture transcriptomics data collected from four patients at Day 0 and Day 6. Each file includes the Ensembl gene ID, the common name of the gene, and the read count of the gene. Since not much metadata is available on the dataset, we assume that culturing and treatment across samples were identical.

There are 5 variables present in the original dataset: patient, culture day, the Ensembl Gene Record and Common Gene Name (redundant variables, for the data analysis, only Ensembl Gene Record is used), and Read Count. All variables are independent and categorical except for read count, which is a dependent continuous variable.

Original Count data is heavily left-skewed with the majority of gene counts between . Performing a log fold correction to the read count shows that the distribution of log counts per gene has a bimodal distribution consistent among all files.

For more information regarding the data structure, distribution, and data cleaning/wrangling, please view the following link. All code is currently in PosterProject.py:
https://github.com/BestAbrar/BINF-5503-Group-5-Poster-Project

Research Questions

For this project, the proposed research questions are:

1. Is there a consistent expression pattern seen after the 6 days of spheroid culturing?
2. Is there a consistent expression pattern seen across all four patient samples?
3. Do the most differentially expressed genes involve related pathways?

Hypotheses

For our research questions, we have developed the following hypotheses:

1) Null hypothesis - there will be no significant change in expression between days 0 and 6 of culture.
   Alternative hypothesis - a significant change in gene expression will be observed after 6 days of spheroid culturing.
2) Null hypothesis - there will be no significant differences in expression between patient samples.
   Alternative hypothesis - gene expression patterns will vary between patient samples.
3) Null hypothesis - differentially expressed genes will share no relation in function or cellular pathway.
   Alternative hypothesis - differentially expressed genes will originate from connected pathways.

Predictions

Given that this dataset includes a massive number of quantified genes, there are several trends which we expect to see in the data. Genes involved in cellular adhesion will likely see increases in expression across all patients, as this family of proteins has been shown to influence the formation and structure of cancer spheroids (Micek et al. 2023; Molika et al. 2004). This could include genes such as FN1, CDH1, and ACTN1. Additionally, HGSOCs are characterized by increased expression of estrogen receptors and decreased expression of progesterone receptors. It therefore seems likely that the genes encoding these receptors (ESR1 / ESR2 and PGR respectively) would show the corresponding shifts in expression (Lisio et al. 2019). Another metric used to differentiate HGSOCs from lower-grade lesions is a Ki-67 proliferation index, which we expect to be elevated in this dataset (Lisio et al. 2019). Cyclin E1 amplification is linked to the chemotherapy resistance of several prominent cancer types, most prominently HGSOCs, making it a likely candidate for differential expression in these patient-derived spheroids (Vias et al. 2023). A 2025 study on epithelial ovarian cancers found a unique behaviour in the typically overexpressed MYC. This study identified that while MYC is indeed upregulated in cancer lesions, upon cellular detachment into spheroids, the expression of MYC decreases as cells enter an acquired dormancy (Buensuceso et al. 2025). We therefore expect to see MYC expression decrease across all patients from day 0 to day 6. Finally, a 2023 study found a correlation between MYC and ZWINT expression, and given the role of ZWINT in proliferation and migration, we expect it to trend with MYC in spheroid development (Vias et al. 2023).

Proposed Methods

Differential gene expression analysis will be used to answer our research questions. This method identifies genes upregulation and downregulation after 6 days of incubation, allowing us to conduct further studies on genes with the most differential expression pattern after incubation. The first step of our analysis involves data cleaning and filtering. The files contain rows of “__no_feature”, “__ambiguous”, “__too_low_aQual”, ”__not_aligned”, “__alignment_not_unique”, which are not representative of the expression data in this project, thus removed before analysis. Genes with 0 read count on both Day 0 and Day 6 across all patients will be filtered out, as well as the genes with consistently low to no expression. Further data cleaning might be required as our analysis proceeds.

In this study, we will be using DESeq 2, limma, or EdgeR for differential gene expression analysis. For data cleaning, processing, and plotting, we will be using pandas, numpy, and seaborn. We plan to incorporate tables, scatter plots, and volcano plots to effectively visualize findings.

There are some anticipated roadblocks in our study. There are only two time points (Day 0 and Day 6) with minimum information in-between, and with only four patients in this study, there is low statistical confidence that the sample population is representative. As previously mentioned, the treatment information on the short-term spheroids culture is unclear, so assumptions have to be made, and it might affect our findings. Another possible challenge would be in determining whether expression patterns are unique to the spheroids or general expression patterns for HGSOC.

References

1. Kim, J., Park, E. Y., Kim, O., Schilder, J. M., Coffey, D. M., Cho, C.-H., & Bast, R. C. (2018). Cell Origins of High-Grade Serous Ovarian Cancer. Cancers, 10(11), 433. https://doi.org/10.3390/cancers10110433
2. Wensink, G.E., Elias, S.G., Mullenders, J. et al. Patient-derived organoids as a predictive biomarker for treatment response in cancer patients. npj Precis. Onc. 5, 30 (2021). https://doi.org/10.1038/s41698-021-00168-1
3. Kordowitzki, P., Lange, B., Elias, K. M., Haigis, M. C., Mechsner, S., Braicu, I. E., & Sehouli, J. (2025). Transforming treatment paradigms: Focus on personalized medicine for high-grade serous ovarian cancer. CA: a cancer journal for clinicians, 75(5), 436–460. https://doi.org/10.3322/caac.70008
4. Lisio, M.-A., Fu, L., Goyeneche, A., Gao, Z.-h., & Telleria, C. (2019). High-Grade Serous Ovarian Cancer: Basic Sciences, Clinical and Therapeutic Standpoints. International Journal of Molecular Sciences, 20(4), 952. https://doi.org/10.3390/ijms20040952
5. Vias, M., Morrill Gavarró, L., Sauer, C. M., Sanders, D. A., Piskorz, A. M., Couturier, D. L., Ballereau, S., Hernando, B., Schneider, M. P., Hall, J., Correia-Martins, F., Markowetz, F., Macintyre, G., & Brenton, J. D. (2023). High-grade serous ovarian carcinoma organoids as models of chromosomal instability. eLife, 12, e83867. https://doi.org/10.7554/eLife.83867
6. Micek, H. M., Rosenstock, L., Ma, Y., Hielsberg, C., Montemorano, L., Gari, M. K., Ponik, S. M., & Kreeger, P. K. (2023). Model of collective detachment in high-grade serous ovarian cancer demonstrates that tumor spheroids produce ECM to support metastatic processes. APL bioengineering, 7(1), 016111. https://doi.org/10.1063/5.0132254
7. Molika, P., Leetanaporn, K., Chiangjong, W., Choochuen, P., & Navakanitworakul, R. (2024). Proteomic Analysis Reveals Cadherin, Actin, and Focal Adhesion Molecule-Mediated Formation of Cervical Cancer Spheroids. Cells, 13(23), 2004. https://doi.org/10.3390/cells13232004
8. Buensuceso, A., Borrelli, M.J., Ramos Valdés, Y. et al. Reversible downregulation of MYC in a spheroid model of metastatic epithelial ovarian cancer. Cancer Gene Ther 32, 83–94 (2025). https://doi.org/10.1038/s41417-024-00850-z